THE CATEGORY OF FILTERED OBJECTS IS ADDITIVE WITH KERNELS AND COKERNELS BUT IS NOT ABELIAN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the first three assertions, each of which is written out with the relevant filtration; the fourth is referred to a named same-chapter example carried on this card as a navigation stub, which is the standard form for a counterexample the author has already recorded elsewhere.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Filtered objects. Same chapter, the definition of a filtered object of an abelian category and of its filtration steps. Supplies the objects, their filtrations and the morphisms whose kernels, cokernels, images and coimages are computed in the proof.
- A counterexample to abelianness. Same chapter, Example example-not-abelian. Supplies the witness for the last assertion, namely that the category of filtered objects is not abelian in general.

Source reference. Original section 0108: sources/stacks-homology/pinned-original-source.tex, source lines 536–536. Navigation pointer invoked at proof line 4951; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. The category of filtered objects $Fil(\mathcal{A})$ has the following properties:*

(1) *It is an additive category.*

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(2) *It has a zero object.*
(3) *It has kernels and cokernels, images and coimages.*
(4) *In general it is not an abelian category.*

Proof. It is clear that $\mathrm{Fil}(\mathcal{A})$ is additive with direct sum given by $(A, F) \oplus (B, F) = (A \oplus B, F)$ where $F^p(A \oplus B) = F^pA \oplus F^pB$. The kernel of a morphism $f : (A, F) \to (B, F)$ of filtered objects is the injection $\mathrm{Ker}(f) \subset A$ where $\mathrm{Ker}(f)$ is endowed with the induced filtration. The cokernel of a morphism $f : A \to B$ of filtered objects is the surjection $B \to \mathrm{Coker}(f)$ where $\mathrm{Coker}(f)$ is endowed with the quotient filtration. Since all kernels and cokernels exist, so do all coimages and images. See Example 0108 for the last statement. $\square$